

CSE Summary

Summary of *Cognitive Systems Engineering: Connecting Situations and Awareness* by John M. Flach — a career-retrospective collection of blog posts tracing the Cognitive Systems Engineering (CSE) field founded by Jens Rasmussen, Erik Hollnagel, Dave Woods, and Don Norman in the 1980s.

What CSE Is

CSE treats the human + technology + work domain as a single unit of analysis — a "joint cognitive system" — rather than separate components to be optimized independently. It emerged because increasing automation pushed humans from direct control into supervisory roles, where they must detect, diagnose, and improvise for situations designers never anticipated.

Information Processing → Meaning Processing

CSE doesn't reject classical human factors (limited working memory, information-processing bottlenecks) — it argues that framework is *insufficient*. Design should not only guard against overload but leverage people's capacity to make sense of situations relative to their goals and to improvise under novelty. This reframes several classic contrasts:

- Workload → situation awareness
- Task analysis (what people do) → work analysis (why, given domain constraints)
- Matching user expectations → shaping them toward smarter strategies

Work Domain Analysis

Built on Rasmussen's Abstraction-Decomposition Space (Purpose → Abstract Function → General Function → Physical Function → Physical Form), which formalizes how experts move between "why" (higher abstraction) and "how" (lower detail) when diagnosing problems or making sense of situations. Crucially, work analysis is a *co-requisite* to design, not a prerequisite — it's never "finished," so it must proceed alongside prototyping ("Serious Play"/co-creation), not before it.

Ecological Interface Design (EID)

Not a visual style but a *process*: uncover a domain's deep structure through Cognitive Work Analysis, then build "configural" displays that show both fine-grained data and higher-order relations simultaneously (both/and, not either/or). The goal is to shape operators' mental models toward better strategies — not just match existing ones — and to support adaptive improvisation rather than rigid procedure-following.

Skilled Muddling

Incrementalism (Lindblom) isn't a weak fallback — done skillfully, it's how experts (and organizations) actually navigate complex, "wicked" problems (Rittel & Webber: no definitive formulation, no stopping rule, one-shot solutions). Flach frames this as a closed loop of affordance (what's possible), specifying (what's perceivable), and satisfying (what matters) — echoing Peirce's abduction. Rational vs. intuitive decision-making is a false dichotomy ("binocular" decision-making — both needed); emotion isn't noise but grounds expertise ("beautiful things work better").

Polycentric / Multi-Loop Control

Organizations and cognition both operate as nested control loops at different timescales (individual: skill-/rule-/knowledge-based; organization: mutual adjustment/planning/standardization), echoed in a reframing of Boyd's OODA loop as three simultaneous layered loops (physical/mental/moral). More degrees of freedom means more resilience but harder control — organizations must tune how tightly outer layers constrain inner ones (rigid Scientific Management vs. loose Mission Command).

The "Myth of Control"

Control theory, Flach argues, isn't really about control — it's about *stability*, an emergent, relational property of a coupled system that can't be attributed to any single component (like a predator-prey balance). This undercuts both the "hero leader" narrative and the "human error as root cause" narrative.

Closing Philosophical Themes

Meaning and experience (even time and temperature) are relational, not purely objective or purely subjective — affordances depend on the coupling between agent and ecology (a frozen pond "supports" differently depending on who's walking on it). Flach ultimately argues meaning is *discovered* through attunement to the world, not purely constructed in the head.

Throughline

CSE is especially relevant now, with AI — the book argues AI's best role in complex domains is enhancing human *observability* and joint sensemaking, not replacing humans as an autonomous controller.